

# IMO(41<sup>st</sup>) – 2000

## First Day

1. AB is tangent to the circles CAMN and NMBD. M lies between C and D on the line CD is parallel to AB. The chords NA and CM meet at P; the chords NB and MD meet at Q. The rays CA and DB meet at E. Prove that PE = QE.
2. A, B, C are positive reals with product 1. Prove that  $\left(A - 1 + \frac{1}{B}\right)\left(B - 1 + \frac{1}{C}\right)\left(C - 1 + \frac{1}{A}\right) \leq 1$ .
3.  $k$  is a positive real. N is an integer greater than 1. N points are placed on a line, not all coincident. A move is carried out as follows. Pick any two points A and B which are not coincident. Suppose that A lies to the right of B. Replace B by another point  $B'$  to the right of A such that  $AB' = kBA$ . For what values of  $k$  can we move the points arbitrarily far to the right by repeated moves?

## Second Day

4. 100 cards are numbered 1 to 100 (each card different) and placed in 3 boxes (atleast one card in each box). How many ways can this be done so that if two boxes are selected and a card is taken each, then the knowledge of their sum alone is always sufficient to identify the third box?
5. Can we find N divisible by just 2000 different primes, so that N divides  $2^N + 1$ ? [N may be divisible by a prime power.]
6.  $A_1A_2A_3$  is an acute angled triangle. The foot of the altitude from  $A_i$  is  $K_i$  and the incircle touches the side opposite  $A_i$  at  $L_i$ . The line  $K_1K_2$  is reflected in the line  $L_1L_2$ . Similarly the line  $K_2K_3$  is reflected in the line  $L_2L_3$  and  $K_3K_1$  is reflected in the line  $L_3L_1$ . Show that the three new lines form a triangle with vertices on the incircle.

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